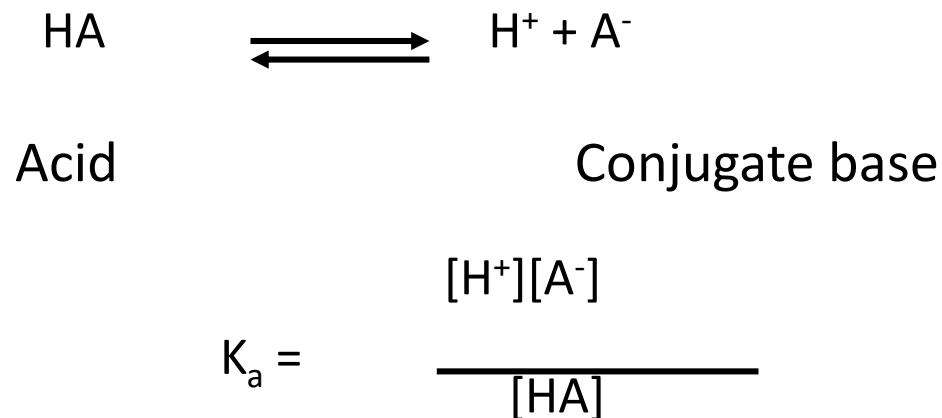


Acids and bases
And
Clinical significance
January
2025

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Acids and bases

- Acids are molecules which can donate a proton (hydrogen ion, H^+)
- Bases are proton acceptors
- Strength of an acid depends on how readily it does that
- Measured by the *acid dissociation constant* K_a



What is pH?

- Measurement of the amount of protons in a solution

$$\text{pH} = -\log_{10}[\text{H}^+]$$

- Logarithm to base 10 of the proton concentration
- Because of the logarithm, a change of one pH unit implies a tenfold change on proton concentration!
- An equivalent measure for acid strength is

$$\text{pK}_a = -\log_{10}[\text{K}_a]$$

pH

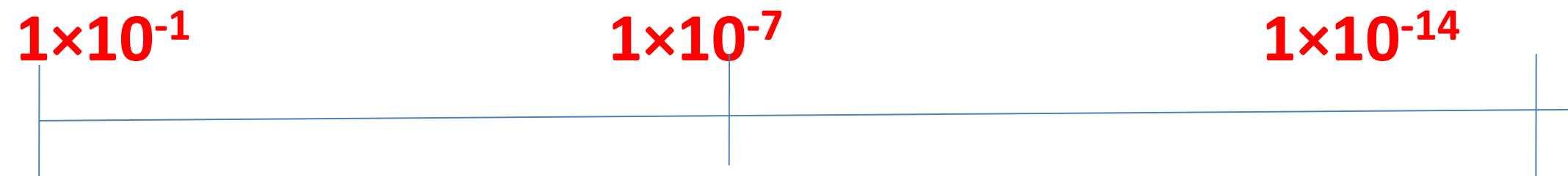
Because hydronium ion concentrations for most solutions are number with negative exponents, these concentrations are more commonly expressed as pH where;

$$\text{pH} = -\log [\text{H}^+]$$

A solution is acidic if its pH is less than 7

A solution is basic if its pH is more than 7

A solution is neutral if its pH is equal to 7



$\text{pH} = -\log [\text{H}^+]$ $\text{pH} = -\log [\text{H}^+]$ $\text{pH} = -\log [\text{H}^+]$

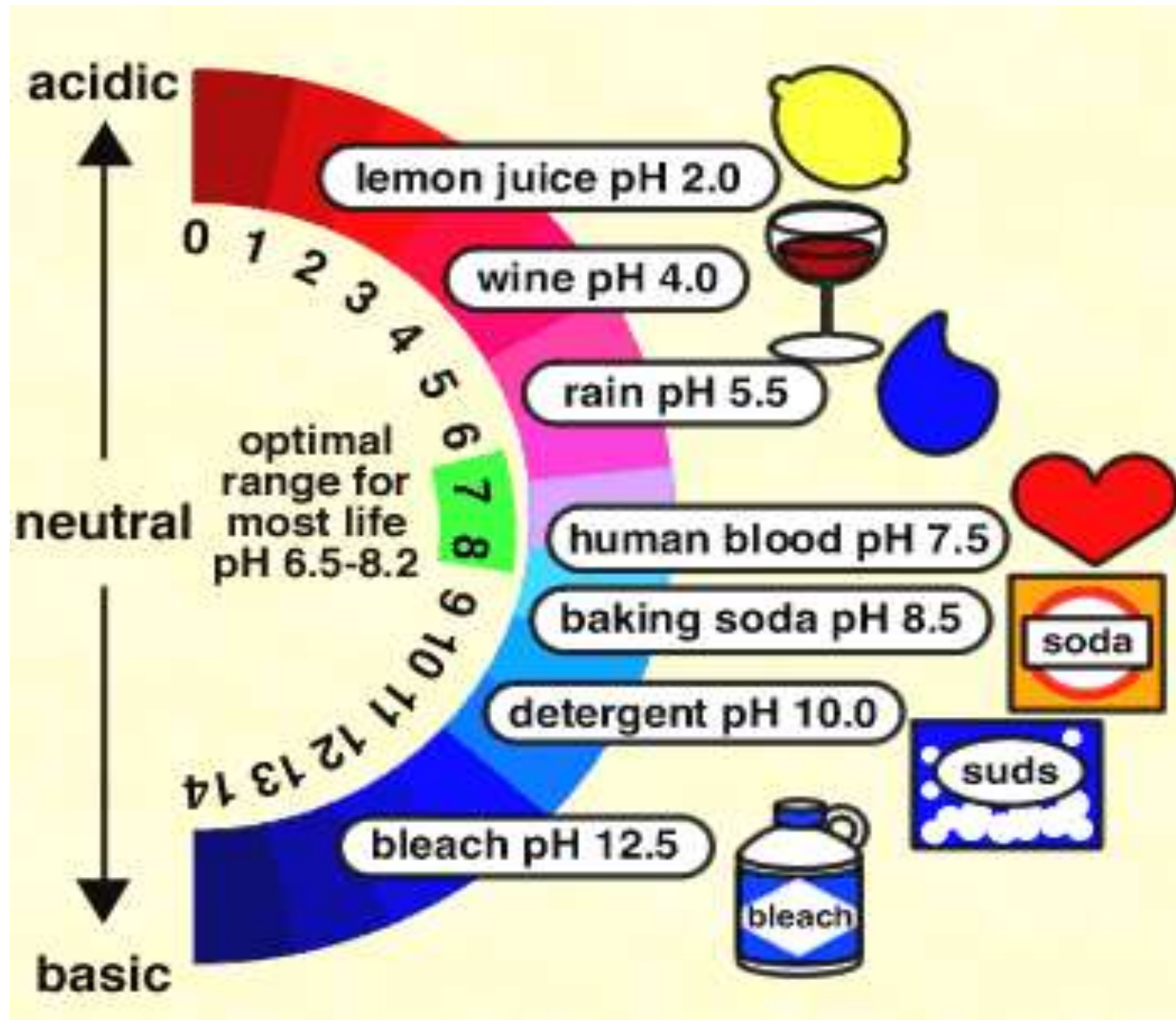
$\text{pH} = -\log [1 \times 10^{-1}]$ $\text{pH} = -\log [1 \times 10^{-7}]$ $\text{pH} = -\log [1 \times 10^{-14}]$

$\text{pH} = 1$

$\text{pH} = 7$

$\text{pH} = 14$

pH Scale



pH paper

- pH paper changes color to indicate a specific pH value.



Buffers

A solution consists of a weak acid and its salt or a weak base and its salt. A solution that resists change in pH when limited amounts of an acid or base are added to it.

pH: The closer the pH of the buffer is to the pK_a of the weak acid the greater the buffer capacity.

Concentration: The greater the concentration of the weak acid and its conjugate base, the greater the buffer capacity.

Blood buffers

The normal range pH of human blood is between (7.35-7.45), any change longer than 0.1 pH unit in either direction may cause illness, if the pH goes below 6.8 or above 7.8 death may result. To hold the pH of the blood close to 7.4, the body uses three buffer systems: carbonate, phosphate, and proteins.

The most important of these systems is the carbonate buffer.

$$\text{pK}_a = 6.37$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$\frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} \text{ ratio is about } 10/1$$

0.025 M HCO₃⁻, and 0.0025 M H₂CO₃

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$\text{pH} = 6.37 + \log \frac{[0.025]}{[0.0025]}$$

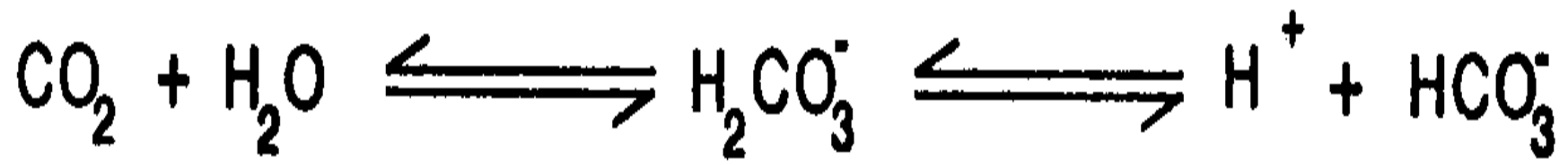
$$\text{pH} = 6.37 + \log 10$$

$$\text{pH} = 6.37 + 1 = 7.37$$

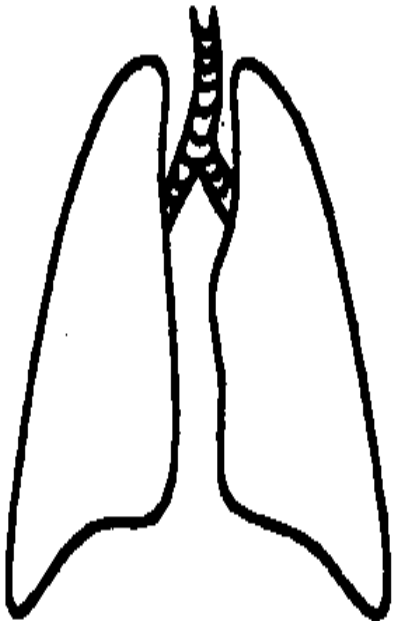
Buffering

- Carbonic acid-bicarbonate buffer system can compensate for an excess or shortage of H⁺

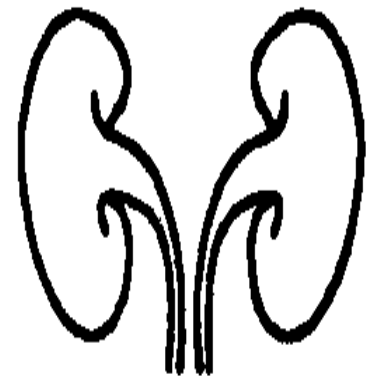




Lungs



Kidneys



Acidosis and alkalosis

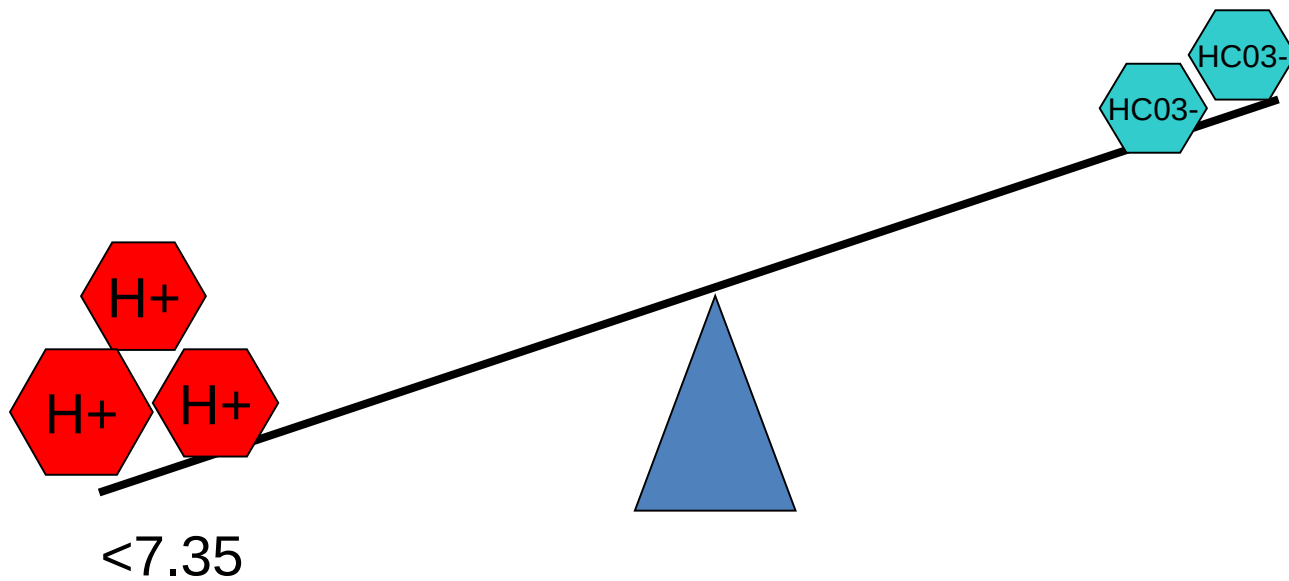
The pH of blood is normally between 7.35-7.45. if the pH goes lower than this level, the condition called acidosis. A blood pH higher than 7.45 produces alkalosis, both are abnormal conditions.

Acidosis leads to depression of the nervous system, mild acidosis can result to fainting, a more sever case can cause coma.

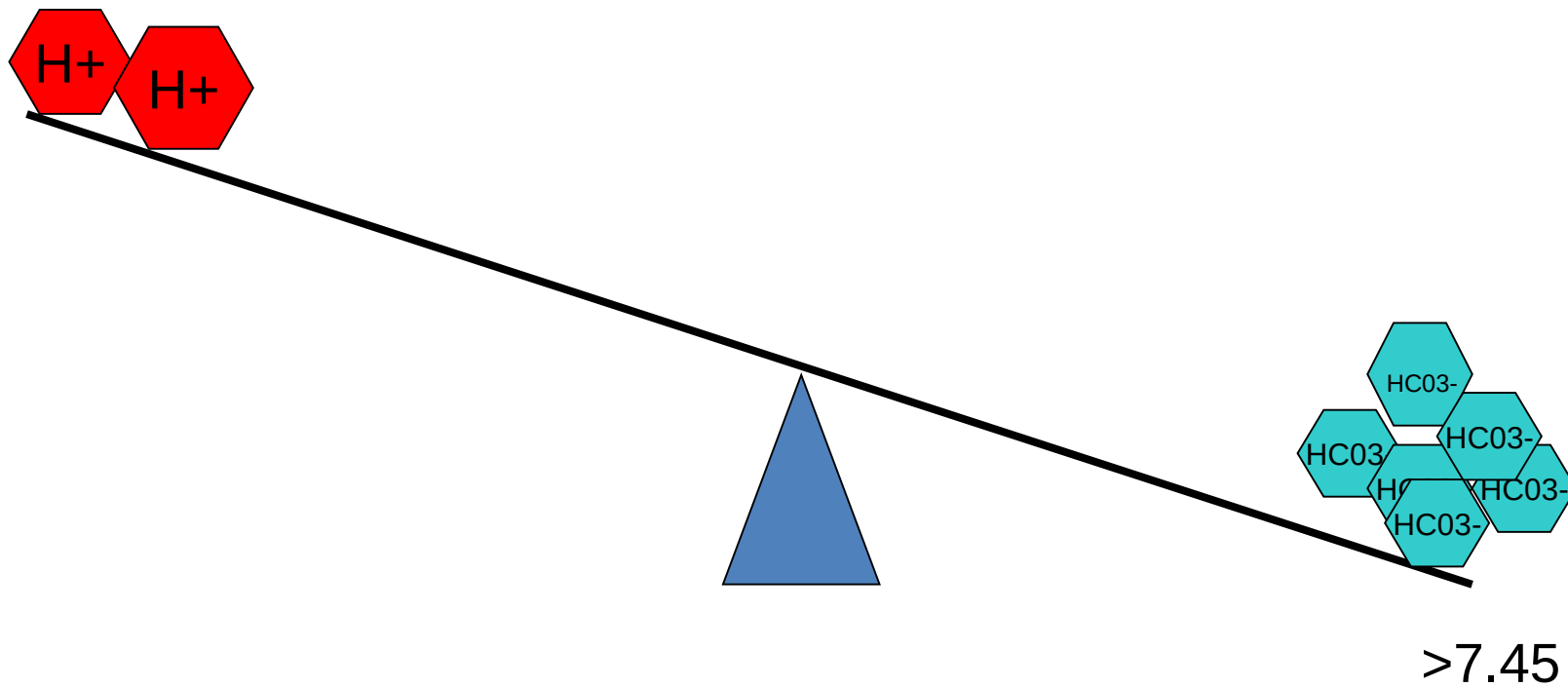
Alkalosis leads to overstimulation of the nervous system, muscle cramps, and convulsions.

If the acidosis or alkalosis persists for a sufficient period of time, or if the pH gets too far away the 7.35-7.45, death may result.

Acidaemia



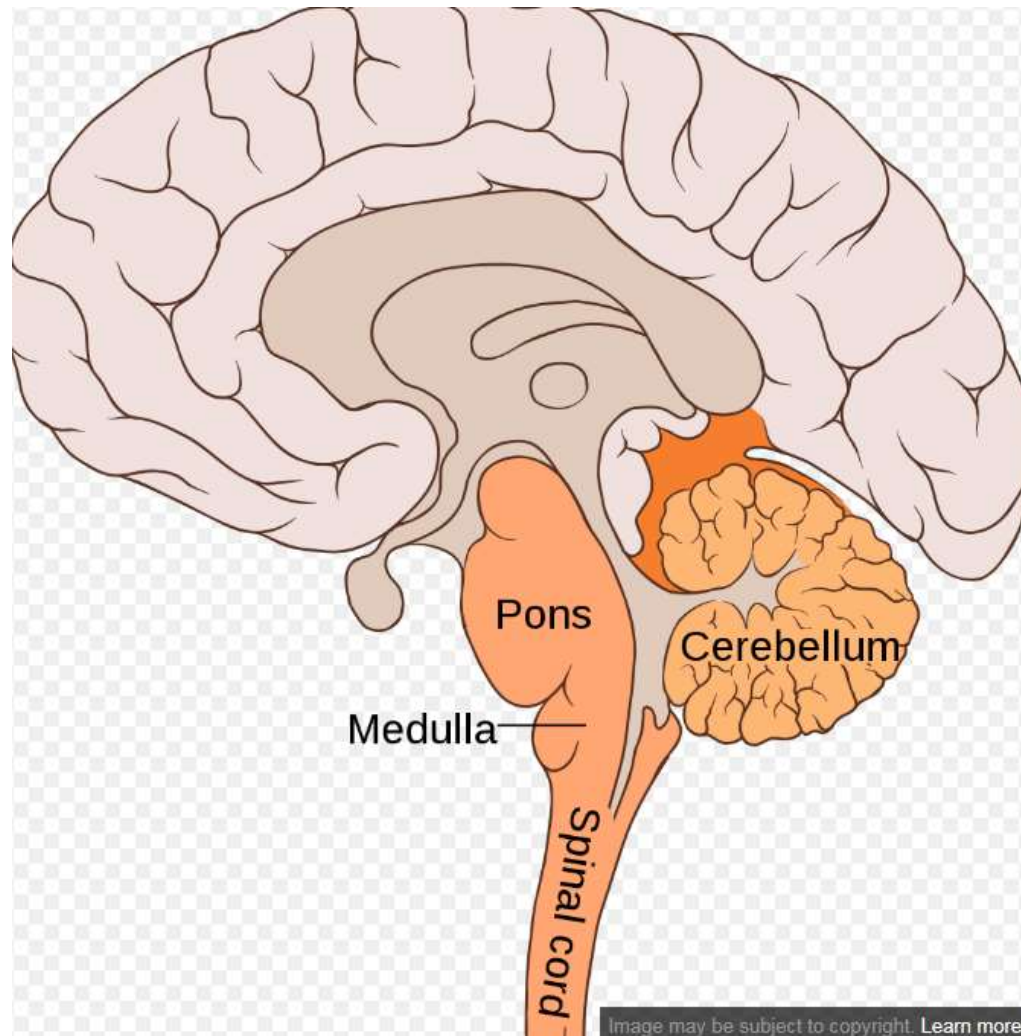
Alkalaemia



Respiratory Regulation

- Respiration is controlled by the Medulla in the brain
- Central Chemo-receptors pick up changes in the concentration of CO₂ and H⁺ in the blood
- High CO₂ or H⁺ will produce deeper more rapid breathing
- Low CO₂ or H⁺ will produce slower more shallow breathing
- **WHY?**

Respiratory Regulation



Metabolic Regulation

- Cells in the distal tubules sense pH change
- The kidneys secrete H^+
- Reabsorb Bicarbonate ions
- Make new Bicarbonate

Respiratory acidosis

Results from difficulty in breathing (hypoventilation). An obstruction in the windpipe or diseases such as pneumonia, emphysema, asthma, or congestive heart failure may diminish the amount of CO₂ that leaves the body through the lungs. (you can produce mild acidosis by holding your breath).

Hypoventilation also can be caused by;

- a. Narcotics
- b. Sedatives
- c. Anesthetics
- d. Depressant drugs

Respiratory acidosis “continuous”

The pH of the blood decreases because the CO₂ unable to escape fast enough, remain in the blood.

where it lowers the $[\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$,



according to Henderson-Hasselblach equation;

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

Respiratory acidosis

pH<7.35, pCO₂>45 Hg mm

- Cause:
 - Respiratory failure
- Symptoms:
 - coma
- Therapy:
 - Therapy of underlying disease
 - Mechanical ventilation

Respiratory alkalosis

Involves the reverse process, it arises from rapid or heavy breathing, called hyperventilation. This condition can come about from fever, infection, the action of certain drugs or even hysteria. Here the excessive loss of CO₂, raises the $[\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$ ratio and the pH.

Respiratory alkalosis

pH>7.45, pCO₂<35 Hgmm

- Causes:
 - Hypoxic respiratory failure
 - Salicylate intoxication
 - Early sepsis
 - Hepatic failure
 - Arteficial hyperventilation
- Symptoms
 - Decreased cerebral or coronary blood flow
 - Neuromuscular excitability
- Therapy:
 - Therapy of underlying disease
 - Increasing the dead space

Acute Respiratory Imbalance

- Respiratory Acidosis

pCO₂ ↑
HCO₃⁻ ↔

- Respiratory Alkalosis

pCO₂ ↓
HCO₃⁻ ↔

Longer Term/Chronic Respiratory Imbalance

- Respiratory Acidosis

pCO₂ ↑

HCO₃⁻ ↑

- Respiratory Alkalosis

pCO₂ ↓

HCO₃⁻ ↓

Metabolic acidosis

The causes of metabolic acidosis are starvation (fasting), and heavy exercise, when the body doesn't get enough food, it burns its own fat, and the products of this reaction are acidic compounds, that enter the blood. This problem sometimes arises in people on fad diets. Heavy exercise causes the muscles to produce excessive amounts of lactate, which makes muscles feel tired and sore.

METABOLIC ACIDOSIS is also caused by a number of metabolic irregularities. For example the disease of diabetes mellitus produces acidic compounds called **ketone bodies**.

Metabolic acidosis

- Symptoms:
 - Increased sympathetic activity
 - Decreased entropy, arterial vasodilatation – critical pH (7,2)
 - Decreased oxy-Hb binding
 - Hyperkalaemia
 - Insulin resistance
 - Free radical formation
 - Bone demineralization
 - Hyperventilation

Metabolic alkalosis

Can result from various metabolic irregularities as well. In addition, it can be caused by excessive vomiting. The contents of the stomach are strongly acidic (pH 1-3), and loss of substantial amounts of this acidic material raises the pH of the blood.

Metabolic alkalosis

- Symptoms
 - Hypoventilation, respiratory depression
 - Neuromuscular excitability
 - Hypokalaemia
 - Seizures
 - Increased oxy-Hgb affinity
 - Altered coronary blood flow, vasoconstriction
 - Decreased cerebral blood flow

Metabolic Imbalance

- Metabolic Acidosis

HCO_3^- ↓
pCO₂ ↓

- Metabolic Alkalosis

HCO_3^- ↑
pCO₂ ↑

Laboratory findings in acid-base disturbances

	pH	pCO ₂	HCO ₃ ⁻	HCO ₃ /H ₂ CO ₃
Normal	7.35-7.45	40mmHg	20mm/Hg	20
Metabolic acidosis	Decreased	Decreased	Decreased	Decreased
Metabolic alkalosis	Increased	Increased	Increased	Increased
Respiratory acidosis	Decreased	Increased	Normal	Decreased
Respiratory alkalosis	Increased	Decreased	Normal	Increased